Exploring the Hiring, Pay, and Trading Patterns of U.S. Firms: The Dominance of Multinationals Engaged in Related-Party Trade

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Abstract

We link U.S. job records with both firm-level business register and customs records to construct a novel set of summary statistics and descriptive regressions that highlight the central role played by the small set of multinational firms (denoted RP XM firms) who engage in both importing and exporting with related parties in translating international trade shocks to shifts in labor demand. We find that RP XM firms 1) dominate trade volumes; 2) account for very disproportionate shares of national employment and payroll; 3) employ greater shares of workers in higher pay deciles; 4) disproportionately poach other firms' high paid workers; 5) offer higher raises to their existing workers. These hiring and pay patterns generally exist even among new RP XM firms, but strengthen with RP XM tenure, and continue to hold, albeit at smaller magnitudes, after conditioning on standard proxies for firm and worker productivity. Taken together, these findings reveal that RP XM status is a reliable proxy for the kind of firm that drives the initial labor market impacts of trade shocks, and that high paid workers are likely to be most directly exposed to such shocks.

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1 Introduction

International trade shocks can have sizable impacts on labor markets, with substantial heterogeneity in earnings and employment impact across workers from different industries and initial pay levels (Autor et al. 2014, Dix-Carneiro and Kovak 2017). However, better understanding how past and future changes in the international trade environment translate from product market shocks to labor market shocks requires assessing the types of firms that are most reliant on international trade and the kinds of workers they hire.

In this paper, we link U.S. job records with both establishment-level data and customs records to construct a novel set of summary statistics and descriptive regressions analyzing the relationship between firms' trading activities and their pay and hiring patterns.

We expand upon Handley et al. (2021), Kamal et al. (2022), and Setzler and Tintelnot (2021) by highlighting the particular need to distinguish multinational firms that both export to and import from related parties (foreign establishments of the same firm) from purely domestic importers and exporters as well as from multinationals whose related-party trade is unidirectional or nonexistent. We show that these particular kind of multinational firms (denoted by RP XM) are central to understanding which types of workers are likely to be directly exposed to trade wars, supply chain disruptions, and other trade shocks that target particular trading activities in specific industries.

Our first set of findings demonstrates that RP XM firms are outliers in both the labor and product markets. Even though only 0.24% of U.S. firms both export and import with related parties in a typical year, such firms account for 20% percent of U.S. employment, 28% percent of payroll, and 42% of revenue. They also disproportionately dominate international goods trade, accounting for 81.3% and 83.5% of U.S. goods exports and imports, respectively. This reflects the fact that they trade much more intensively than firms that only import or export outside the firm (i.e. at "arms-length"). We also show that RP XM firms generally persist in related-party importing and exporting in subsequent years as well, while firms that only import or export often stop doing so or add the other activity. Collectively, these properties make RP XM status a reliable proxy for the kind of firm that drives the initial labor market impacts of trade shocks.

Our second set of findings reveals that annual earnings distributions at RP XM firms are quite left-skewed even relative to other importing and exporting firms, with considerably smaller shares of low-paid workers and much higher shares in the top U.S. earnings decile. This pattern holds in nearly every industry sector. We then show that RP XM status predicts over $8,000 in increased annual earnings relative to other importing and exporting firms even when controlling flexibly for combinations of industry and firm revenue category and for firms' intensity of importing and exporting. This suggests either that related-party trade captures an additional dimension of firm productivity beyond

revenue and trade volume, or that related-party trade itself may directly affect firms' hiring and pay patterns.

Furthermore, by comparing specifications with and without worker fixed effects, we show that about 90% of RP XM, firms' excess pay per worker is due to superior worker which still leaves these firms paying a premium of about $750 on average and nearly $2,500 among the most export-intensive $RP\,XM$ firms.

In a third set of findings, we explore the mechanisms through which high-paid workers become concentrated at RP XM firms. We show that these firms disproportionately poach workers who were already paid in the highest decile, particularly those previously employed at other RP XM firms. To attract and retain such workers, RP XM firms offer new hires substantially higher pay than their previous positions and give existing employees substantially larger raises. Both sources of pay premia are particularly concentrated among the most export-intensive and experienced RP XM firms. This suggests that a very small group of highly productive multinationals are fiercely competing for the most talented workers, which in turn makes high-paid workers particularly exposed to trade shocks.

Taken together, our findings imply that a tiny and relatively consistent group of firms disproportionately mediate the overall and distributional impact of trade shocks on the labor market. These insights demonstrate the need for quantitative models of trade that combine separate treatment of multinationals with worker skill heterogeneity, a la Setzler and Tintelnot (2021). They also illustrate the value of collecting related-party information as part of customs data in other countries.

Our paper relates most directly to three literatures. First, the mid-2000s arrival of firm-level administrative data spawned a wave of descriptive analyses examining the characteristics of exporting and importing firms (see, e.g., Bernard et al. 2007 and Bernard et al. 2012). These papers emphasize that manufacturing trading firms (particularly importers and exporters) tend to be larger, higher-paying, and more productive, and that a small share of firms accounts for a large share of total import and export volume. We contribute by showing that, even relative to firms that both import and export, the very few who do both activities with related parties are extreme in their employment, revenue, and reliance on international trade, and are quite stable over time. Furthermore, we show that RP XM firms display these patterns across a wide variety of industries beyond manufacturing.

Both Setzler and Tintelnot (2021) and Kamal et al. (2022) also document that multinationals typically feature greater employment, revenue, and average pay than purely domestic firms. However, they identify multinationals using foreign business tax payments rather than related-party trade transactions. Our trade-based classification allows us to compare RP XM firms with arms-length importers and exporters, and to account

for and assess the role of trade intensity and trade experience. Furthermore, we examine differences in pay across the entire distribution.

A second literature explores how and why trading firms exhibit higher average pay. Several theoretical papers emphasize that high-productivity firms are more likely to engage in international trade in general and related-party trade in particular (Melitz 2003, Halpern et al. 2015, Antras and Helpman 2004). Others hypothesize and provide evidence that more productive firms also benefit more from high-ability workers, and that labor market search frictions lead them to pay a premium to find their desired workers faster (e.g. Helpman et al. 2010, Lentz and Mortensen 2010, and Card et al. 2018). We show that even among trading firms, RP XM firms are distinct in both worker composition and pay premia, consistent with Setzler and Tintelnot (2021)'s comparisons of multinationals and non-multinationals. Moreover, we document that these firms' greater pay premia reflect both greater pay increases for new hires relative to these workers' outside options as well as higher raises.

Several papers in this literature have sought to establish that engaging in exporting causes higher pay (Araújo and Paz 2014, Macis and Schivardi 2016, Brambilla et al. 2017), perhaps because it generates additional rents that are partially shared with workers. Frías et al. (2022) show in the Mexican context that this causal effect operates via changes in both worker composition and pay premia for a given set of workers. While we do not seek to establish causality, our regression results are consistent with a causal effect on worker hiring and pay from establishing related- party trading relationships. In particular, we show that the differences in worker composition and pay premia extend to importing firms, are larger at importing and exporting firms, and are largest at RP XM firms.

Finally, a third literature spanning several subfields highlights the growing power of a select group of "superstar" or "mega" firms. These firms have been classified as outliers using a variety of characteristics: employment, markups, productivity, worker composition, firm pay premia, digital capital, and rates of innovation (Autor et al. 2020, Song et al. 2019, Tambe et al. 2020, Braguinsky et al. 2023). Our paper demonstrates that these firms are also distinguished by their consistent willingness to export and import with their own foreign establishments.

2 Data

We construct a database that combines three large, restricted, administrative datasets from the U.S. Census Bureau: the Longitudinal Business Database (LBD), the Longitudinal Firm Trade Transactions Database (LFTTD), and the Longitudinal Employer-Household Dynamics database (LEHD). The LBD leverages tax records from the universe of U.S. employing establishments to provide establishment-level data on annual employ-

ment, payroll, and industry. It also contains firm-level revenue for most firms. The LFTTD consists of customs transaction data for all goods imports and exports indicating the importing or exporting U.S. firm, the value of the goods transacted, and whether the transaction was with a different firm (at arms length) or a foreign establishment of the same firm (related party). The LEHD reports quarterly earnings for each job (worker-firm) match for a sample of 25 states that approved our project.

We use this database to construct a firm-level sample and a worker-level sample. The firm-level sample reflects all U.S. employing firms contained in the firm-level LFTTD-LBD merge from 2007-2017. The worker-level sample consists of a 30% random sample of the set of workers who appear in our 25-state LEHD in some year between 2007-2014 and whose firms are linked to the firm-level sample ($\sim$ 138M worker-years). This sample provides us with workers' detailed earnings and employment histories combined with their firms' characteristics from the LBD-LFTTD.

3 Trade Exposure Measures and Regression Specifications

We consider three ways to classify firms according to their trading patterns. Our primary classification, "trade engagement status", assigns each firm-year to one of the following categories: non-trading (NT), importer only (arms-length and/or related-party, denoted M), exporter only (arms-length and/or related-party, denoted X), exporter and importer with at most unidirectional related-party transactions (XM), and related-party importer and related-party exporter (RP XM). XM includes firms who import and export exclusively at arms-length along with firms whose imports *or* exports include related-party trade, but not both.[1] Regardless of how such firms are classified, one cannot cleanly compare RP XM and non-RP XM multinationals in our data, since multinationals that do not trade goods with their foreign affiliates cannot be distinguished from purely domestic firms.

We distinguish importers from exporters because exporting generally increases labor demand through increased output, while importing affects labor demand both through a similar scale effect (via lower unit costs) and a substitution effect that may reduce employment. Furthermore, exporting may alter worker composition by requiring employees who can arrange shipments and manage multiple product lines, while importing may cause offshoring of the production staff (Hummels et al., 2018). RP XM status may compound these effects by increasing firm productivity, requiring a broader set of managers to coordinate trade with foreign affiliates (Gumpert et al., 2021), and reorganizing supply

[1] Unidirectional related-party traders exhibit pay patterns that more closely resemble XM firms than RP XM firms.

chains in ways that may increase or decrease resilience to various trade disruptions.

Second, we also classify firms according to the total number of previous years in which they also featured the current year's status (0, 1, 2-3, or 4+). This allows us to analyze the persistence of firms' trade engagement and the rate at which the pay patterns of firms who change status converge to the norm in their new status.

Finally, our trade engagement categories focus on the extensive margin of whether a firm performs a particular trade activity, but are silent about how intensively they perform these activities. The extent of firms' reliance on international trade is likely to determine the sensitivity of its costs and product demand, and therefore labor demand, to international trade shocks. We measure import and export "intensity" via the ratio of a firm's total imports or exports to its total revenue, and assign import and export intensity quartiles based on these ratios' employment-weighted national distributions.

When analyzing firms' pay distributions, we assign each worker-year to one "dominant" firm that accounted for the largest share of the worker's earnings. To approximate full-year pay for workers who work fewer than four quarters at their dominant firms, we annualize earnings by multiplying by four their average quarterly earnings among "full" quarters that are preceded and followed by positive earnings.

We initially construct basic summary statistics to characterize the joint distribution of firms' pay policies and measures of trade engagement. These unconditional relationships are useful for forming prior beliefs about which workers are likely to be sensitive to trade shocks. This is because the kinds of workers who disproportionately work at the most heavily trading firms will, at least initially, be more exposed to trade shocks even if firms' trade activity and worker composition are jointly driven by other underlying factors such as firm productivity.

We then use several worker-level regression specifications to explore whether these unconditional relationships hold conditional on other firm and worker characteristics. Controlling for firm characteristics reveals whether trade engagement measures are valuable predictors of firms' labor market behavior above and beyond standard proxies for firm productivity and industry, while controlling for worker characteristics distinguishes differences in worker quality from composition-adjusted pay premia.

Let $j(i,t)$ capture worker i's dominant firm in year t. Our baseline regression specification is:

$$Y_{it} = \sum_{e} 1(Trade\ Engagement_{j(i,t)t} = e)\beta^{e} + \mathbf{D}_{j(i,t)t}^{n \cdot f_s}\boldsymbol{\omega}^{n \cdot f_s} + \epsilon_{it} \tag{1}$$

We initially use annualized earnings as the outcome Y_{it}, but we also consider earnings growth between $t-1$ and t as well as indicators for whether worker i's earnings fall into a given national earnings decile.

$1(Trade\ Engagement_{j(i,t)t} = e)$ selects the worker's trade engagement status e, while β^e captures the outcome change associated with working in a firm with status e relative to the omitted category, working in a non-trading firm. $\mathbf{D}^{n \cdot fs}_{j(i,t),t}$ represents a design matrix that selects combinations of 4-digit NAICS industry and firm size decile, with $\boldsymbol{\omega}^{n \cdot fs}$ capturing the corresponding industry-by-size fixed effects. Firms are assigned to national employment-weighted size deciles based on employment or revenue, depending on specification. ϵ_{it} captures remaining determinants of Y_{it}, including measurement error.

Augmented specifications add worker fixed effects or initial earnings decile fixed effects (for the earnings growth outcome) or replace the trade engagement status indicators with indicators for combinations of trade engagement status and either categories of status experience or import or export intensity quartiles.

4 Results

4.1 The Firm-Level Distributions of Trade Engagement, Intensity, and Experience

Figure 1 shows a remarkable concentration of the U.S. economy among the mere 0.24% of firms engaging in related-party exporting and importing (panel 1). These few firms manage to generate 20% of U.S. employment (panel 2), 28% of payroll (panel 3), and 48% of revenue (panel 4). Furthermore, they also dominate U.S. international trade, accounting for 81.4% of all exports and 83.5% of imports. Adding in the other categories of active trade engagement, we find that trading firms collectively hire 53% workers, pay 62. 6% of all earnings and generate 73. 4% of all revenues, consistent with Handley et al. (2021). Figure 1 highlights that both the organizational structure (arms-length versus related-party) and the type of trade activity (imports, exports, or both) matter for the interplay between trade and the labor market.

Figure 2, which replicates Figure 1 for each NAICS supersector, illustrates that there exists substantial heterogeneity across sectors in the shares of firms, employment, payroll, and revenue accounted for by each trade engagement status. The manufacturing sector, the trade, transportation and utilities sector, and the information sector are more likely to participate in international trade generally and related-party trade in particular. However, even the less trade-oriented sectors, such as education & health and leisure & hospitality, still have a substantial share of their workforce, payroll, and revenue in trading firms.

Row 1 of Table 1 displays various quantiles of the worker-weighted distribution of export intensity and import intensity among all exporting firms (combining X, XM, and $RPXM$) and importing firms (M, XM, and $RPXM$), while rows 2-5 provide separate

quantiles by trade engagement category.

Even though a large share of X and M employment occurs in firms that rely very little on international trade, $RP\,XM$ employment is much more concentrated among intensively trading firms. For example, the 50th, 75th, and 95th export intensity quantiles are 0.9%, 8.0%, and 30.6% for RP XM firms versus 0.1%, 1.0%, and 12.8% for X firms. For import intensity, the 50th, 75th, and 95th quantiles are 3.4%, 10.9% and 36.8% for RP XM firms versus 0.04%, 0.03%, and 5.8% for M firms.

Focusing on sector heterogeneity, the remaining rows of Table 1 show that the bulk of employment among exporters and importers within each supersector occurs at firms with low reliance on trade, with the notable exceptions of manufacturing, natural resources (for exporting) and trade/transportation/utilities (for importing). 25% of manufacturing employment occurs at firms with export intensity greater than 17%, and another 25% occurs at firms with export intensity between 7.5% and 17%. These quantiles are 3.4% and 13% for natural resources. For import intensity, 25% and 50% of manufacturing employment occurs at firms featuring import intensity greater than 14.6% and 4.6%, respectively. However, nearly all supersectors feature at least 5% of exporter employment among firms where exports account for 10% of revenue (finance/real estate and education & health are the exceptions). The same is true for import intensity.

Figure 3 plots the shares of U.S. firms, employment, payroll, and revenue accounted for by each combination of trade engagement status and national export intensity quartile (left panels) or import intensity quartile (right panels). The first noticeable pattern is that, regardless of the outcome, the left and right graphs look quite similar to each other. This is because import intensity and export intensity are highly correlated among XM and $RP\,XM$ firms that engage in both activities, so that there is substantial overlap in the set of firms being classified into a given quartile within these trade engagement categories.

The second panel on each side shows that M, X, and XM firms account for most of the employment in the bottom quartile of export and import intensity, while $RP\,XM$ firms account for most of the employment in the top two quartiles.

The last two panels show that this contrast becomes even stronger when shares of national payroll and particularly national revenue are plotted rather than shares of national employment, so that most of the revenue generated by XM firms is concentrated among lower intensity firms, while a striking share of $RP\,XM$ firms' revenue is concentrated among the top quartile of export or import intensity. In fact, 23.2% and 23.8% of national revenue is generated by $RP\,XM$ firms in the top export and import intensity quartiles. Taken together, the panels of Figure 3 suggest that even among the small pool of $RP\,XM$ firms, it is the even smaller subset that is most reliant on trade that particularly dominates both labor and product markets.

Figure 4 explores the persistence and transition patterns of firms' trade activities across years. The leftmost bars in each panel show the share of employment in each trade engagement status category within firms who retain their current status the following year. 91.2% of NT and 87.9% of RP XM employment is concentrated at status-retaining firms, suggesting that large non-trading firms rarely transition to engaging in international trade, and more interestingly, large RP XM firms nearly always continue to engage in intrafirm importing and exporting year after year. By contrast, M, X, and XM statuses are much volatile, with status-retaining firms accounting for only 58.3%, 46.9% and 69.1% of employment.

More generally, moving across the remaining of the first panel, we see that trade status changes become less frequent as the firm gains experience within its current form(s) of trade activity. This increased stability is particularly pronounced among RP XM firms, so that 95% of employment at firms with 4+ years of RP XM experience is concentrated among firms that continue to export and import with related parties.

This stability implies that firms with 4 or more years of RP XM experience account for a very large share (82.5%) of RP XM employment (second panel). In contrast, firms with 4 or more years of experience within their current trade engagement status only account for 35.9%, 20.7%, and 32.1% of employment in the M, X, and XM categories, respectively.

Among firms who alter their trade activities, the third panel provides for each origin trade engagement status the employment-weighted breakdown of the following year's trade engagement status. We see that most firms that begin importing and exporting with related parties were already engaging in both activities at arms length in the previous year. Only 3.7% of all XM firms transition to $RP\,XM$ status, but these firms are disproportionately large, so they account for nearly half of the XM employment at status-changing firms. By contrast, the bulk of the remaining 40-50% of M and X employment that exists at firms that change status is roughly equally split between firms that stop trading internationally altogether and firms that begin to both export and import, with almost none of the initial M or X employment concentrated at firms transitioning to $RP\,XM$.

$RP\,XM$ firms' size, persistence, and reliance on (and dominance of) international trade ensure that their responses to trade shocks will primarily determine how such shocks affect labor demand. Note, though, that if these firms' hiring and pay policies closely resemble those of non-traders and arms-length imports and/or exporters, then their disproportionate exposure to trade shocks would minimally impact these shocks' labor market incidence. However, the next section documents that $RP\,XM$ firms in fact exhibit pay distributions and hiring/retention patterns that differ dramatically even from other trading firms.

4.2 The Relationship between Firm's Trade Activities and Pay Distributions

The top left panel of Figure 5 plots the share of workers within each decile of the national earnings distribution by trade engagement status. The concentration of employment within high-paying deciles is particularly striking for $RPXM$ firms: only 6.6% of their workers come from the bottom national decile, while top decile workers are particularly overrepresented (17%). Other kinds of trading firms generally have very slightly increasing shares as one moves toward higher deciles, while NT firms have by far the least favorable earnings distribution, with employment shares that strictly decrease in earnings decile.

The other panels of Figure 5 plot the annual earnings distribution by trade engagement for each supersector. These panels show that the $RPXM$ earnings distribution generally dominates the earnings distributions of all other categories of trade engagement regardless of industry.

Table 2 displays the estimated annual earnings premium associated with working in trade engaged firms. It shows that these patterns generally hold once we control for standard proxies for firm productivity such as employment, revenue, and industry via the regression approach described in Section 3. Specifically, workers in $RPXM$ firms earn \$20,590 more than NT workers even after controlling for fixed effects for combinations of 4-digit NAICS industry and worker-weighted employment-based firm size decile (column 1). The conditional earnings gap relative to NT firms is smaller at trading firms with other forms of trade engagement but still sizable: \$3,647 at X firms, \$6,540 at M firms, and \$10,240 at XM firms, respectively.

Column 2 of Table 2 uses revenue rather than employment to assign firms to size deciles when constructing the NAICS $\times$ size controls. The results are qualitatively similar but with significantly smaller magnitudes, with the adjusted earnings gap for $RPXM$ vs. NT falling by 58% to \$8,648 and those of the other trading statuses falling by between 70% and 90%. This finding suggests that trade engagement may be capturing a similar dimension of firm productivity as revenue, and thus may be a useful proxy in contexts where trade data is available but revenue data is not. Nonetheless, $RPXM$ status in particular still strongly predicts firm pay even conditional on revenue, highlighting the importance of collecting data that distinguishes multinationals from other firms.

The earnings coefficients in the first two columns reflect both the tendency of $RPXM$ firms to hire high-skilled workers as well as the premium they pay conditional on skill relative to workers' outside options. To isolate the pay premia, in columns 3 and 4 of Table 2 we control for worker composition by including worker fixed effects. Regardless of whether employment-based or revenue-based size controls are used, coefficients on

each trade engagement status generally fall by around 90% when worker fixed effects are added. Thus, disproportionate hiring of high-productivity workers accounts for the lion's share of trading firms' elevated pay per worker. However, $RPXM$ firms do seem to pay meaningfully more than their workers' outside options, with annual earnings premia of \$2,443 and \$748 when employment-based and revenue-based size controls are used, respectively. In particular, the last row shows that earnings premia at $RPXM$ firms are statistically significantly higher than at XM firms, even though both types of firms export and import.

Next, we explore how these conditional earnings premia contribute to the distributional differences in pay across trade engagement categories. Figure 6 displays each trade status's effect on the probability that a worker's earnings falls into each national earnings decile, conditional on other firm and worker characteristics. Using employment-based firm size controls, $RPXM$ status predicts that a firm will feature a significantly more left-skewed earnings distribution relative to NT or any other status. For example, it increases a worker's probability of being in the top earnings decile by 8.4 percentage points and reduces the probability of being paid in the bottom decile by 5.0pp relative to working in a NT firm. More generally, workers at trading firms are less likely to fall into each below-median decile and more likely to fall into each above-median decile, with coefficients that grow larger in magnitude at the two ends of the distribution, so that the corresponding impacts on top (bottom) decile probabilities for XM, X, and M firms are 3.8pp (-3.5pp), 1.3pp (-1.6pp), and 2.4pp (-2.2pp), respectively.

Adding worker fixed effects does not change the shape of the pattern or the ordering of trade statuses, but dramatically reduces the magnitudes, indicating that differences in skill composition explain the bulk of the predictive power of trade engagement status throughout the earnings distribution. Nonetheless, with employment-based size controls, working in a $RPXM$ firm still predicts a 3.8% (-2.3%) change in the probability of being paid in the top (bottom) decile, even when compared to that worker's samplewide average. The bottom two panels of Figure 6 repeat this exercise when firm size deciles are assigned based on firm revenue rather than employment. We still observe that workers are more likely to be paid above the national median when working at any type of trading firm, and that $RPXM$ firms are particularly unlikely to pay in the bottom couple of deciles. In particular, working at an $RPXM$ firm still decreases the probability of landing in the bottom decile of pay by 1.5% when worker fixed effects are excluded and 0.7% when they are included.

We next examine whether greater trade intensity might also contributes to more generous pay even conditional on trade engagement status. Panel A of Figure 7 displays the unconditional distributions of earnings deciles by export intensity quartile within each trade engagement status. Among $RPXM$ firms, greater intensity is quite clearly

associated with a shift toward higher pay deciles, with almost a quarter of employees at the most export intensive $RP\,XM$ firms being paid in the top national earnings decile compared to 9% in the 5th decile and a mere 2% in the bottom decile. By contrast, the earnings decile distribution is fairly similar across export intensity quartiles for both X firms and XM firms, with higher intensity predicting a slightly *less* favorable distribution. The same pattern emerges for import intensity (Panel B): larger shares of high paid employees at $RP\,XM$ firms with greater import intensity, and smaller shares of high paid employees at more import intensive M and XM firms.

Columns 1 and 2 of Table 3 confirms that these findings generally hold when industry $\times$ revenue-based firm size controls are introduced. In particular, when regressing annual earnings on interactions between trade status dummies and export intensity quartile dummies, we see that the positive relationship between export intensity and earnings is quite weak among X firms, stronger among XM firms, and extremely powerful for $RP\,XM$ firms: earnings are predicted to be more than \$10,000 higher for RP XM workers at top export quartile firms than at bottom quartile firms, compared to gaps of $\sim$ \$5,000 and \$1,000 for XM and X workers, respectively. The results for importing intensity (columns 5 and 6) are qualitatively similar but slightly more muted, with milder and less consistent increases in predicted pay with greater intensity. In both panels, we also see clear, ascending pay gaps within each export (or import) intensity quartile as one moves from X (or M) to XM to $RP\,XM$ firms. Thus, both the unconditional and conditional results indicate that the nature and intensity of trade both predict firm pay distributions, and that intensity more strongly predicts pay at $RP\,XM$ firms than at other trading firms.

Consistent with the trade engagement results that pool intensities, adding worker fixed effects (columns 3-4 and 7-8 of Table 3) reduces pay gaps by status and by intensity quartile within status to around 10-20% of their original size, suggesting that both sources of pay differences are primarily driven by differences in the kinds of workers hired by such firms rather than pay policies conditional on worker skill. However, pay premia at $RP\,XM$ firms in the top export intensity quartile remain quantitatively important - an additional \$2,500-\$3,000 in annual earnings relative to a worker's samplewide average when compared to either bottom quartile $RP\,XM$ firms or X firms.

Figure 8 shows that by and large, the earnings distributions are quite similar across experience categories within a trade engagement status. This result suggests that either firms' pay distributions already resembled those of their new trading status before they transition or that their pay distributions converge very quickly to the norm within their new status.

That said, there are subtle changes in pay with greater status-specific experience. $RP\,XM$ firms with 4+ years of experience, who have a proven ability to operate success-

fully as multinationals, have the most left-skewed pay distributions. On the other hand, XM firms' pay distributions become less generous with experience, perhaps because firms with high XM experience have revealed that they are unlikely to be productive enough to become full $RPXM$ multinationals.

To assess sensitivity to industry×size controls, Table 4 displays results from a version of our baseline specification in which separate earnings coefficients are estimated for each experience category within each firm trade engagement status.[2] Even firms that have just achieved RP XM, status already pay $2,735 more than non-trading firms after accounting for the effects of industry and revenue-based firm size. Additional experience as $RPXM$ predicts substantial further increases in pay per employee of $6,577 after one year, $7,983 after 2-3 years, and $10,054 after four or more years. Column 2, which adds worker fixed effects, reveals that the high pay of firms new to RP XM status is purely a reflection of the quality of workers they hire, since new RP XM firms' annual earnings premia is actually lower than those of either new or experienced non-trading firms. Instead, the status' higher overall higher pay premium is driven by an estimated $1,149 premium for firms with 4+ years of RP XM experience.

Our worker fixed effect results suggest that trading firms generally and $RPXM$ firms in particular attract and retain a disproportionate share of workers who command high pay regardless of their firm. The following section explores the mechanisms by which they do so.

4.3 How Do Multinational Firms Attract High Quality Workers?

Figure 9 displays the shares of new hires by each trade engagement category whose pay in the prior year fell into each decile of the earnings distribution. Job movers' prior year earnings tend to be low regardless of destination, since low pay partly explains their willingness to incur job switching/search costs. However, new hires by trading firms and particularly $RPXM$ firms show a smaller degree of negative selection on prior pay, and $RPXM$ firms show a unique propensity to attract workers that were already being paid in the top national decile.

Figure 10 further disaggregates Figure 9 by displaying the same prior year decile shares among new hires by combination of origin and destination trade engagement status. The figure also displays shares of each destination status' hires who came from each origin trade engagement status. Even when restricting to hires previously at $RPXM$

[2]We focus on the separate experience coefficients among RP XM firms, since firms first entering RP XM status are unambiguously deepening their trade activities by either importing or exporting with a related party for the first time, while firms first entering other trading categories may have either added or removed a trade activity, making their coefficients harder to interpret.

firms, $RPXM$ destinations disproportionately attract workers from the top decile. By contrast, job movers from NT, M, and X actually tend to exhibit lower prior earnings among those going to $RPXM$ firms than to other trade statuses. This suggests that $RPXM$ firms are primarily targeting initially high paid workers with $RPXM$ experience. Indeed, $RPXM$ firms' disproportionate tendency to hire workers from other $RPXM$ firms accounts for much of their overall ability to poach existing high paid workers, since, as just demonstrated, the origin $RPXM$ firms also tend to pay high earnings premia, particularly to their highest paid workers. In particular, workers coming from $RPXM$ firms constitute 41.8% of $RPXM$ hires, but no more than 21.4% of new hires for any other status. More generally, each trade status hires a disproportionate share of workers from other firms of the same trade status. This suggests that trading firms may value workers who have experience in coordinating relevant trade activities, though the pattern could also be driven by match effects on other firm characteristics correlated with trade engagement status, such as having experience in the same industry as the hiring firm.

Figure 11 reinforces this point by presenting coefficients from augmented earnings regressions that allow the impact of the hiring firm's trade engagement status to depend on the status of the worker's origin firm. The thin bars in the top panel shows that new hires tend to enjoy higher pay if they were already working at another firm with the same trade engagement status. This is particularly true of $RPXM$ firms, who pay \$14,144 more to their hires from $RPXM$ firms compared to those from XM firms even after accounting for revenue/industry differences. While this primarily reflects their selective poaching of the high-paid workers from such firms workers the pattern remains, albeit at much lower magnitudes, when worker fixed effects are included (wider bars). This suggests a willingness by RP XM firms to pay a premium for existing RP XM workers above and beyond these workers' outside options, consistent $RPXM$ firms disproportionately valuing workers with prior multinational experience.

The cross-sectional pay premia paid by trading firms reflects a blend of initial pay for new hires along with higher raises to keep valued employees after they arrive. To see this, Table 5 examines worker earnings growth rather than annual earnings. Columns 1 and 2 report average percentage premiums paid to incumbents at firms of each trade engagement status relative to the raises experienced by stayers at NT firms using employment-based and revenue-based size controls, respectively, while columns 3 and 4 report corresponding pay premia among new hires relative to NT firms' new hires. All growth specifications control for workers' initial earnings deciles in addition to the size decile $\times$ NAICS fixed effects.

While workers who switch jobs generally enjoy much greater earnings growth than job stayers regardless of trade status, both stayers and new hires' earnings grow faster

at trading firms than at NT firms. However, as before, the premium is about 1.5 times as large at $RP\,XM$ firms as at their XM counterparts (2.1% vs. 1.3% extra growth for stayers, 16% vs. 10% for new hires), which constitutes a statistically significantly difference in both cases. The qualitative patterns are the same when revenue-based size deciles are used (columns 2 and 4 of Table 5), but are generally only around one-tenth to one-fifth as large, again reflecting the strong relationship between firm revenue and trade engagement status. Even with flexible controls for revenue category, though, incumbents and new hires at $RP\,XM$ firms experience 0.67% and 6.5% higher earnings growth than their respective counterparts at NT firms.

Figure 12 shows that $RP\,XM$ firms' larger raises do in fact translate to lower turnover rates: 19.5% of $RP\,XM$ workers are replaced each year compared to 21.3% for XM workers and 26.8% for NT workers. Looking across the initial pay distribution, we see that $RP\,XM$ turnover rates are particularly low among their highest paid workers (13.7%), consistent with the evidence that their pay policies are particularly geared toward retaining their most skilled workers.

Along the same lines, the second panel of Figure 11 shows that workers who leave RP XM firms tend to experience disproportionately large percentage earnings losses unless they get hired by another RP XM firm. This reinforces the idea that these workers have context-specific experience or skill that is not fully utilized at non-multinational firms, incentivizing them to stay within the $RP\,XM$ sector once they join it.

Columns 3-4 and 7-8 of Table 3 shows growth premia for both new hires and stayers for combinations of trade status and trade intensity. Conditional on either export or import intensity, both raises for job stayers and earnings increases for new hires are consistently higher at $RP\,XM$ firms than either X or XM. Among workers hired by $RP\,XM$ firms, those joining firms in the top export quartile receive a 5-7% larger initial earnings increase than those joining firms at lower intensity quartiles. Then such workers experience subsequent annual earnings growth that is 0.2%-0.5% faster. Both gaps are reversed in sign for X, while for XM the earnings growth gaps are somewhat smaller for new hires and similarly sized for job stayers. Few systematic differences in earnings growth by import intensity within trade engagement status exist for either hires or stayers across trade statuses, suggesting that export intensity is a stronger predictor of pay policies than import intensity.

Columns 3 and 4 of Table 4 display growth premia for stayers and new hires for combinations of trade status and experience within a status instead. Among firms with 4 or more years of experience within their current trade engagement status, $RP\,XM$ firms exhibit pay growth premia for retained workers and for new hires that are at least 0.3 and 3 percentage points higher than any other trade status.

Among $RP\,XM$ firms, growth premia increase monotonically from with greater ex-

perience, and are more than twice as large at the most experienced firms as at those first gaining RP XM, status (4.5% vs. 2.2%). The results for stayers show greater pay raises for $RP\,XM$ firms with 4 or more years of experience (1%) than those with 0 or 2-3 years of experience (around 0.2%), although we find a noisily estimated 1.5% pay raise premia among the small set of workers at $RP\,XM$ firms with 1 year of experience.

Synthesizing the earnings growth results from Tables 3-5, we find that among $RP\,XM$ firms, the types that have particularly left-skewed pay distributions generally lure new hires with considerably higher salaries than they previously received and induce them to stay with larger raises. Thus, the combination of long-lived $RP\,XM$ status and high trading volume as a share of revenue may be a particularly accurate proxy for underlying firm productivity and for complementarity with worker skill. Overall, our results strongly suggest it is the highest paid workers whose earnings are most exposed to international trade shocks, since they receive substantial pay premia that are concentrated among this small set of very large $RP\,XM$ firms.

5 Conclusion

In this paper, we leverage linked data combining firm-level customs and revenue information with job-level earnings and personnel data to investigate the relationship between firms' trade activities and their hiring and pay practices. While we confirm that firms who engage in international trade generally employ more workers, earn more total revenue, and pay more per worker, we particularly focus on the degree to which multinational firms whose U.S. establishments engage in within-firm importing and exporting with their foreign affiliates exhibit dramatically different profiles of both trade intensity and pay distributions, even relative to firms that engage in both importing and exporting at arms length. Across a broad swath of industries, we show that these related-party importers and exporters are far more reliant on international trade for both their inputs and their sales than other trading firms, and that they are far more likely to employ the highest paid workers and far less likely to employ the lowest paid workers in the U.S. economy. This primarily reflects a greater tendency to attract and retain the highest skilled workers. However, we also find that these firms pay both higher initial salaries and larger raises relative to what these same workers received at other firms, which may be a key mechanism by which they attract top talent. These relationships continue to hold, albeit with smaller magnitudes, even when we control flexibly for industry, total employment, and even total revenue. This suggests that these trade engagement classifications are not merely duplicative proxies for unobserved total factor productivity, but instead provide independent predictive power. In addition, We show that firms who become related-party importers and exporters tend to remain so, and that their pay patterns very quickly con-

verge to multinationals who have greater experience with such activities. These findings demonstrate the importance of obtaining information on the organizational structure of firms' trade, which is rarely collected outside the U.S.

Taken together, our results suggest that 1) these multinationals' responses to international trade shocks mediate the shocks' labor market impact, and 2) the workers whose jobs are most immediately impacted by these adjustments are disproportionately highly paid. Thus, a complete understanding of the contribution of trade globalization to inequality and the incidence of particular trade shocks requires a greater focus on the responses of multinational firms. In particular, the equilibrium labor market impact of trade shocks also depends on the flexibility of $RPXM$ firms' supply chains, their ability to offshore particular kinds of workers, and the portability of those workers' skills to other job opportunities. We pursue the latter issue in parallel work (Carballo and Mansfield (2024)).

References

Antras, Pol and Elhanan Helpman, "Global sourcing," *Journal of political Economy*, 2004, *112* (3), 552–580.

Araújo, Bruno César and Lourenço S. Paz, "The effects of exporting on wages: An evaluation using the 1999 Brazilian exchange rate devaluation," *Journal of Development Economics*, 2014, *111*, 1–16. Special Issue: Imbalances in Economic Development.

Autor, David, David Dorn, Lawrence F Katz, Christina Patterson, and John Van Reenen, "The fall of the labor share and the rise of superstar firms," *The Quarterly Journal of Economics*, 2020, *135* (2), 645–709.

Autor, David H, David Dorn, Gordon H Hanson, and Jae Song, "Trade Adjustment: Worker-level Evidence," *The Quarterly Journal of Economics*, 2014, *129* (4), 1799–1860.

Bernard, Andrew B., J. Bradford Jensen, Stephen J. Redding, and Peter K. Schott, "Firms in International Trade," *Journal of Economic Perspectives*, September 2007, *21* (3), 105–130.

_ , _ , _ , and _ , "The Empirics of Firm Heterogeneity and International Trade," *Annual Review of Economics*, 2012, *4* (1), 283–313.

Braguinsky, Serguey, Joonkyu Choi, Yuheng Ding, Karam Jo, and Seula Kim, "Mega Firms and Recent Trends in the US Innovation: Empirical Evidence from the US Patent Data," Technical Report, National Bureau of Economic Research 2023.

Brambilla, Irene, Nicolas Depetris Chauvin, and Guido Porto, "Examining the Export Wage Premium in Developing Countries," *Review of International Economics*, 2017, *25* (3), 447–475.

Carballo, Jeronimo and Richard Mansfield, "What Drives the Labor Market Incidence of Trade Shocks?: An Equilibrium Matching Analysis of China's WTO Accession," Technical Report 2024.

Card, David, Ana Rute Cardoso, Joerg Heining, and Patrick Kline, "Firms and labor market inequality: Evidence and some theory," *Journal of Labor Economics*, 2018, *36* (S1), S13–S70.

Dix-Carneiro, Rafael and Brian K. Kovak, "Trade Liberalization and Regional Dynamics," *American Economic Review*, October 2017, *107* (10), 2908–46.

Frías, Judith A., David S. Kaplan, Eric Verhoogen, and David Alfaro-Serrano, "Exports and Wage Premia: Evidence from Mexican Employer-Employee Data," *The Review of Economics and Statistics*, 03 2022, pp. 1–45.

Gumpert, Anna, Henrike Steimer, and Manfred Antoni, "Firm Organization with Multiple Establishments*," *The Quarterly Journal of Economics*, 12 2021, *137* (2), 1091–1138.

Halpern, László, Miklós Koren, and Adam Szeidl, "Imported inputs and productivity," *American Economic Review*, 2015, *105* (12), 3660–3703.

Handley, Kyle, Fariha Kamal, Wei Ouyang et al., "A long view of employment growth and firm dynamics in the United States: Importers vs. exporters vs. non-traders," Technical Report 2021.

Helpman, Elhanan, Oleg Itskhoki, and Stephen Redding, "Inequality and unemployment in a global economy," *Econometrica*, 2010, *78* (4), 1239–1283.

Hummels, David, Jakob R Munch, and Chong Xiang, "Offshoring and labor markets," *Journal of Economic Literature*, 2018, *56* (3), 981–1028.

Kamal, Fariha, Jessica McCloskey, and Wei Ouyang, "Multinational Firms in the U.S. Economy: Insights from Newly Integrated Microdata," Working Papers 22-39, Center for Economic Studies, U.S. Census Bureau September 2022.

Lentz, Rasmus and Dale T Mortensen, "Labor market models of worker and firm heterogeneity," *Annu. Rev. Econ.*, 2010, *2* (1), 577–602.

Macis, Mario and Fabiano Schivardi, "Exports and wages: Rent sharing, workforce composition, or returns to skills?," *Journal of Labor Economics*, 2016, *34* (4), 945–978.

Melitz, Marc J, "The impact of trade on intra-industry reallocations and aggregate industry productivity," *econometrica*, 2003, *71* (6), 1695–1725.

Setzler, Bradley and Felix Tintelnot, "The Effects of Foreign Multinationals on Workers and Firms in the United States*," *The Quarterly Journal of Economics*, 05 2021, *136* (3), 1943–1991.

Song, Jae, David J Price, Fatih Guvenen, Nicholas Bloom, and Till Von Wachter, "Firming up inequality," *The Quarterly journal of economics*, 2019, *134* (1), 1–50.

Tambe, Prasanna, Lorin Hitt, Daniel Rock, and Erik Brynjolfsson, "Digital capital and superstar firms," Technical Report, National Bureau of Economic Research 2020.

6 Tables

Table 1: Quantiles of the Employment-Weighted Distributions of Export and Import Intensity: Overall, by Trade Engagement Category, and by Industry

	Export Intensity					Import Intensity				
	25	**50**	**75**	**95**	**99.5**	**25**	**50**	**75**	**95**	**99.5**
Overall	0.0002	0.0021	0.0294	0.2452	0.6609	0.0003	0.0040	0.0475	0.2901	0.6795
Within Trade Engagement Status										
X	0.0002	0.0012	0.0099	0.1275	0.6845					
M						0.0001	0.0004	0.0028	0.0576	0.4956
XM	0.0000	0.0003	0.0036	0.0962	0.5562	0.0001	0.0007	0.0077	0.1456	0.6144
$RPXM$	0.0011	0.0088	0.0802	0.3059	0.6630	0.0042	0.0335	0.1088	0.3680	0.7146
Within Industry										
Natural Resources	0.0040	0.0341	0.1296	0.3301	0.7796	0.0010	0.0078	0.0417	0.1441	0.3869
Construction	0.0003	0.0018	0.0107	0.1593	0.4739	0.0003	0.0012	0.0066	0.1076	0.4138
Manufacturing	0.0209	0.0756	0.1701	0.4189	0.7709	0.0102	0.0460	0.1460	0.3598	0.6948
Trade, Transp., Util.	0.0001	0.0019	0.0103	0.1484	0.5703	0.0029	0.0312	0.0735	0.3552	0.7192
Information	0.0003	0.0013	0.0095	0.1180	0.4555	0.0007	0.0022	0.0285	0.1333	0.4312
Finance & Real Estate	0.0001	0.0005	0.0080	0.0683	0.4882	0.0000	0.0002	0.0023	0.0673	0.4549
Professional Services	0.0001	0.0010	0.0242	0.2623	0.6413	0.0001	0.0010	0.0291	0.2871	0.6347
Education & Health	0.0000	0.0001	0.0009	0.0224	0.3720	0.0001	0.0002	0.0006	0.0048	0.4375
Leisure & Hospitality	0.0001	0.0009	0.0081	0.1139	0.4574	0.0002	0.0013	0.0130	0.1848	0.3706
Other Services	0.0011	0.0077	0.0403	0.3728	0.7134	0.0006	0.0041	0.0342	0.3028	0.6162

Source: LBD-Rev/LFTTD 2007-2017.

Notes: The left panel of Table 1 displays chosen quantiles of the employment-weighted distribution of export intensity, defined as the firm's total value of exports divided by its total revenue) among all exporting firms, by trade engagement status among exporting firms, and by industry for exporting firms. The right panel displays the corresponding quantiles of employment-weighted import intensity, defined as the firm's total value of imports divided by its revenue. See Figure 1 for definitions of trade engagement categories.

Table 2: Conditional Annual Earnings Gaps for Each Trade Engagement Status Relative to Workers at Non-Trading Firms

	(1)	(2)	(3)	(4)
X	$3,647♦	$397	$440♦	-$51
	(421)	(275)	(95)	(103)
M	$6,540♦	$2,143♦	$825♦	$191♦
	(344)	(306)	(81)	(63)
XM	$10,240♦	$3,118♦	$1,310♦	$340♦
	(430)	(407)	(153)	(107)
RP XM	$20,590♦	$8,648♦	$2,443♦	$748♦
	(771)	(703)	(204)	(201)
Firm Size Dec.	Emp.	Rev.	Emp.	Rev.
FE NAICS4 x Size	X	X	X	X
FE Worker			X	X
R-Squared	0.2677	0.3629	0.9228	0.6438
N (millions)	138	138	138	138
Test:	$10,348♦	$5,530♦	$1,134♦	$409▲
RP XM − XM	(689)	(581)	(157)	(165)

Source: LBD-Rev/LFTTD 2007-2017.
Notes:
♦: p<0.01; ▲: p<0.05; △: p<0.1. Column (1) of Table 2 displays the estimated residual annual earnings gap among workers in each trade engagement status relative to those at non-trading firms after controlling for indicators capturing combinations of firm employment decile and industry supersector. Column (2) replaces the firm employment deciles with firm revenue deciles when forming the interactive indicators. Columns (3) and (4) display residual earnings gaps after controlling for worker fixed effects in addition to employment decile × industry and revenue decile × industry indicators, respectively.

Table 3: Worker Earnings Premiums by Combination of Trade Engagement Status and Import/Export Intensity

		Export Quartiles				Import Quartiles			
		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Quartile			**Stayers**	**New Hires**			**Stayers**	**New Hires**
X	**1st**	-$9	-$272	0.33%▲	0.76%				
		(674)	(331)	(0.16)	(1.2)				
	2nd	$39	$128	-0.065%	0.056%				
		(370)	(101)	(0.13)	(0.7)				
	3rd	$645	-$80	-0.015%	0.051%				
		(420)	(103)	(0.088)	(0.54)				
	4th	$1,175♦	$181▲	0.12%	-0.52%				
		(273)	(91)	(0.08)	(0.57)				
M	**1st**					$2,401♦	$190△	0.32%♦	2.3%♦
						(615)	(105)	(0.12)	(0.66)
	2nd					$1,790♦	$76	0.19%▲	2.5%♦
						(291)	(77)	(0.085)	(0.51)
	3rd					$1,476♦	$337♦	0.54%♦	3.4%♦
						(264)	(93)	(0.12)	(0.55)
	4th					$3,312♦	$570♦	0.62%♦	3.4%♦
						(551)	(127)	(0.15)	(0.79)
XM	**1st**	$1,614♦	$95	0.17%	2.1%▲	$2,160♦	$254	0.16%	2.3%♦
		(598)	(157)	(0.14)	(0.82)	(639)	(165)	(0.13)	(0.8)
	2nd	$1,700♦	$69	0.29%△	2.5%♦	$3,204♦	$378♦	0.27%▲	2.8%♦
		(554)	(161)	(0.15)	(0.78)	(523)	(145)	(0.12)	(0.66)
	3rd	$3,429♦	$486♦	0.39%♦	4%♦	$3,147♦	$312△	0.67%♦	4.9%♦
		(485)	(133)	(0.11)	(0.65)	(482)	(160)	(0.15)	(1)
	4th	$6,677♦	$1,138♦	0.82%♦	6.2%♦	$4,848♦	$575♦	0.43%▲	4.2%♦
		(453)	(168)	(0.11)	(0.57)	(542)	(176)	(1.4)	(0.2)
RP XM	**1st**	$4,294♦	-$677△	5%♦	0.74%	$8,077♦	-$237	6.1%♦	0.72%▲
		(1,368)	(347)	(1.6)	(0.54)	(1,728)	(402)	(0.36)	(1.6)
	2nd	$2,794♦	$358	0.39%	3.2%▲	$6,021♦	$534	0.38%△	5.8%♦
		(1,050)	(332)	(0.35)	(1.3)	(1,053)	(339)	(0.2)	(1.3)
	3rd	$6,389♦	$647△	0.46%	4.7%♦	$7,310♦	$772♦	0.96%♦	6%♦
		(978)	(338)	(0.28)	(1.2)	(880)	(297)	(0.22)	(1.1)
	4th	$14,850♦	$2,478♦	0.9%♦	10%♦	$11,640♦	$1,766♦	0.61%▲	7.4%♦
		(1,021)	(380)	(0.16)	(1)	(1,031)	(292)	(0.24)	(1.1)
Firm Size Dec.		Rev.	Rev.	Rev.		Rev.	Rev.	Rev.	
FE NAICS4 x Size		X	X	X		X	X	X	
FE Worker			X				X		
FE Earn. Dec.				X				X	
R-Squared		0.1608	0.6438	0.2131		0.1608	0.6438	0.2131	
N (millions)		138	138	138		138	138	138	

Source: LEHD/LFTTD 2007-2014

Notes:

♦: p<0.01; ▲: p<0.05; △: p<0.1. Column (1) of Table 3 displays the estimated residual gap in expected annual percentage growth of earnings by combination of trade engagement status and export intensity quartile after controlling for firm revenue decile × industry supersector fixed effects. Column (2) also residualizes with respect to worker fixed effects. Columns (3) and (4) consider earnings growth among workers retained by or newly hired by their current firm as their outcomes, respectively, and control for worker's prior year earnings decile in addition to firm revenue decile × industry supersector fixed effects. The coefficients from columns (3) and (4) are generated by the same earnings growth regression. Columns (5)-(8) mimic (1)-(4) except that import intensity quartiles are used in place of export intensity quartiles.

Table 4: Worker Earnings Premiums by Firm Cumulative Experience within Each Trade Engagement Status

		(1)	(2)	(3)	(4)
	Years Prior			**Stayers**	**New Hires**
NT	**0**	\$1,379♦	-\$139	0.59%♦	3.6%♦
		(413)	(119)	(0.18)	(0.21)
	1	-\$163	-\$708♦	0.31%♦	2.3%♦
		(222)	(65)	(0.088)	(0.13)
	2 – 3	-\$253	-\$685♦	0.46%♦	0.76%♦
		(168)	(36)	(0.052)	(0.13)
	4+				
X	**0**	\$1,049▲	-\$234	0.29%▲	0.44%
		(486)	(156)	(0.15)	(0.41)
	1	\$1,341♦	-\$421♦	0.5%♦	1.2%♦
		(515)	(125)	(0.16)	(0.44)
	2 – 3	-\$112	-\$773♦	0.14%	0.57%
		(503)	(189)	(0.18)	(0.38)
	4+	-\$1,177△	\$183	0.08%	-0.087%
		(629)	(186)	(0.17)	(0.52)
M	**0**	\$2,741♦	-\$202△	0.74%♦	2.3%♦
		(520)	(112)	(0.16)	(0.27)
	1	\$2,500♦	-\$737♦	0.23%	2.2%♦
		(640)	(137)	(0.16)	(0.42)
	2 – 3	\$2,157♦	-\$410♦	0.18%	1.5%♦
		(511)	(149)	(0.13)	(0.36)
	4+	\$1,283▲	\$380▲	0.68%♦	1.4%♦
		(566)	(163)	(0.16)	(0.36)
XM	**0**	\$4,153♦	-\$467♦	0.56%♦	2.5%♦
		(603)	(152)	(0.14)	(0.36)
	1	\$4,413♦	-\$660♦	0.43%▲	2.6%♦
		(577)	(183)	(0.17)	(0.35)
	2 – 3	\$3,150♦	-\$449♦	0.54%♦	2.6%♦
		(551)	(140)	(0.14)	(0.43)
	4+	\$1,385△	\$506♦	0.71%♦	1.5%♦
		(752)	(174)	(0.17)	(0.52)
RP XM	**0**	\$2,735△	-\$827▲	0.17%	2.2%♦
		(1,538)	(349)	(0.44)	(0.7)
	1	\$6,577♦	-\$820▲	1.5%▲	2.9%♦
		(1,362)	(360)	(0.61)	(0.69)
	2 – 3	\$7,983♦	-\$543△	0.2%	3.9%♦
		(1,026)	(294)	(0.32)	(0.65)
	4+	\$10,540♦	\$1,149♦	1%♦	4.5%♦
		(979)	(307)	(0.19)	(0.53)

Source: LEHD/LFTTD 2007-2014

Notes: ♦: p<0.01; ▲: p<0.05; △: p<0.1. Column (1) of Table 4 displays the estimated residual gap in expected annual percentage growth of earnings by category of firm cumulative experience within each trade engagement status after controlling for firm revenue decile × industry supersector fixed effects. Column (2) also residualizes with respect to worker fixed effects. Columns (3) and (4) consider earnings growth among workers retained by or newly hired by their current firm as their outcomes, respectively, and control for worker's prior year earnings decile in addition to firm revenue decile × industry supersector fixed effects. Columns (3) and (4) report coefficients from the same earnings growth regression.

Table 5: Conditional Gaps in Earnings Growth for Each Trade Engagement Status Relative to Workers at Non-Trading Firms.

	(1) Stayers	(2) Stayers	(3) New Hires	(4) New Hires
X	0.54%♦	0.07%	4.2%♦	0.24%
	(0.08)	(0.07)	(0.56)	(0.49)
M	1.0%♦	0.35%♦	7.6%♦	2.7%♦
	(0.06)	(0.07)	(0.34)	(0.37)
XM	1.3%♦	0.35%♦	10%♦	3.3%♦
	(0.07)	(0.08)	(0.48)	(0.52)
RP XM	2.1%♦	0.67%♦	16%♦	6.5%♦
	(0.14)	(0.13)	(0.84)	(0.83)
Firm Size Dec.	Emp.	Rev.	Emp.	Rev.
FE NAICS4 x Size	X	X	X	X
FE Earn. Dec.	X	X	X	X
R-Squared	0.278	0.285	0.278	0.285
N (millions)	138	138	138	138
Test:	0.79%♦	0.32%▲	6.1%♦	3.2%♦
RP XM − XM	(0.13)	(0.12)	(0.72)	(0.72)

Source: LBD-Rev/LFTTD 2007-2017.
Notes:
♦: p<0.01; ▲: p<0.05; △: p<0.1. Column (1) of Table 2 displays the estimated residual gap in expected annual percentage growth of earnings among retained workers (job stayers) in each trade engagement status relative to relative to retained workers at non-trading firms after controlling for indicators capturing combinations of firm employment decile and industry supersector as well as indicators for the worker's earnings decile in the baseline year. Column (2) uses firm revenue deciles rather than employment deciles when forming industry interaction indicators. Columns (3) and (4) mimic columns (1) and (2) but consider earnings growth among workers newly hired workers in each trade engagement status relative to nearly hired workers at non-trading firms. The coefficients from columns (1) and (3) are generated by the same earnings growth regression, as are the coefficients from columns (2) and (4).

7 Figures

Figure 1: The Shares of Firms, Employment, Payroll, and Revenue in Each Trade Engagement Category



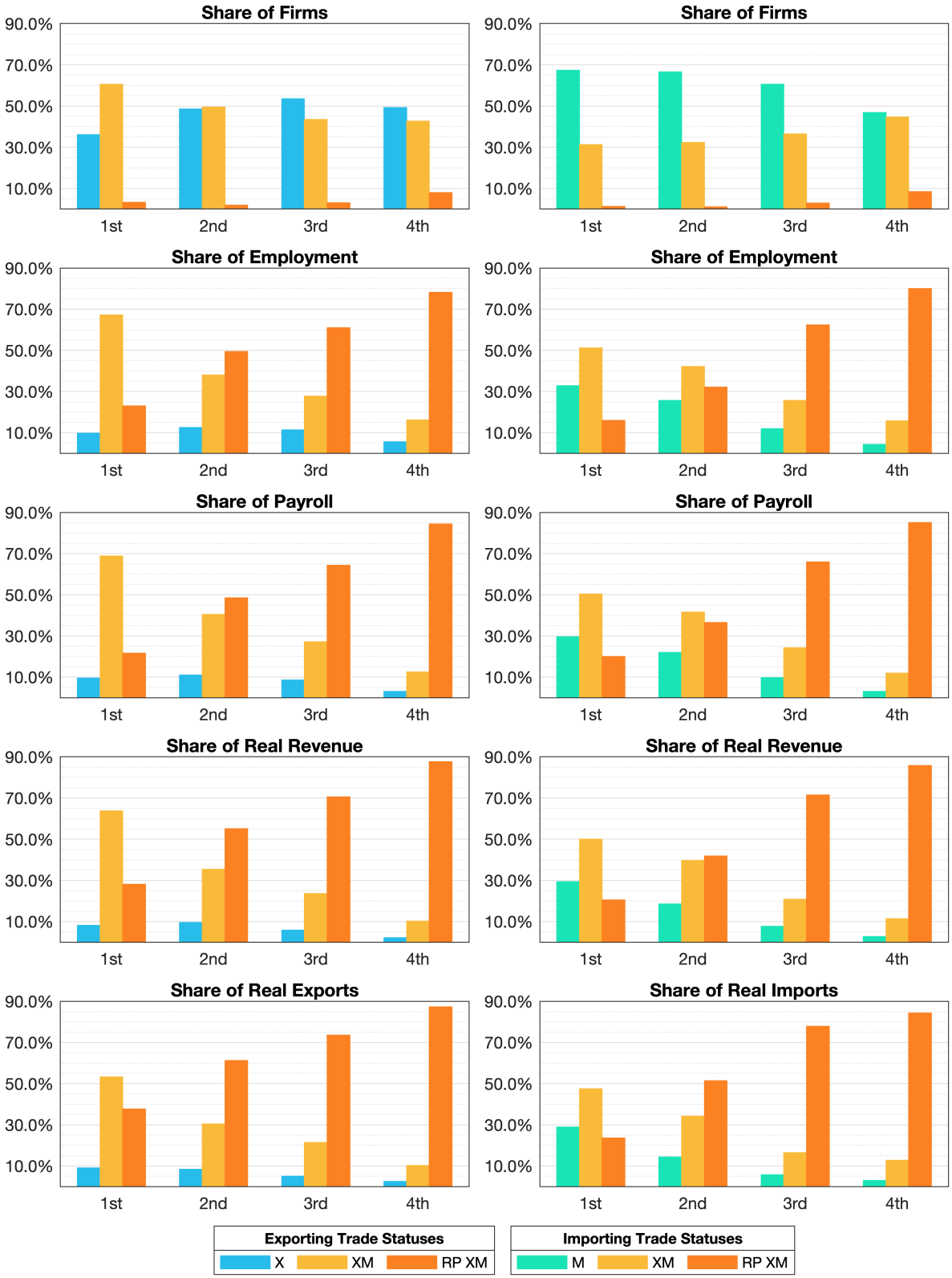
Source: LBD-Rev/LFTTD 2007-2017.
Notes: Panels a)-f) of Figure 1 depict the trade engagement status composition of U.S. firms, total employment, total payroll, total revenue, total goods imports, and total goods exports, respectively. "XM": Firms that exhibit positive values for both imports and exports in the LFFTD in the chosen year, but at least one of these activities is only performed using arms-length transactions. "$RP\,XM$: Firms that record positive values of related-party imports and related-party exports.

Figure 2: The Shares of Firms, Employment, Payroll, and Revenue in each Trade Engagement Category by Industry Supersectors



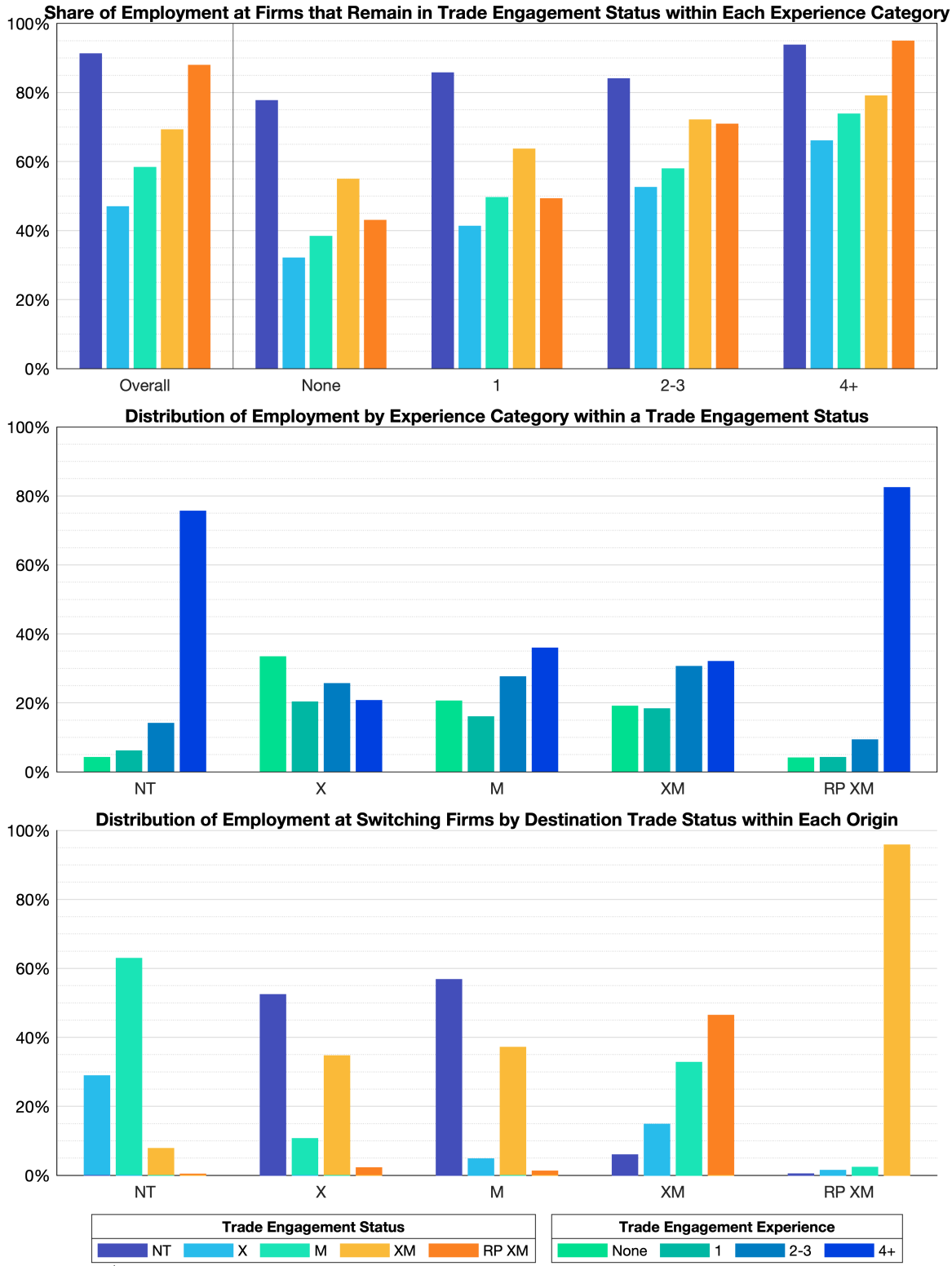
Source: LBD-Rev/LFTTD 2007-2017.
Notes: Panels a)-f) of Figure 2 display the trade engagement status composition of U.S. firms, employment, payroll,revenue, imports, and exports by industry supersector. The share of each outcome accounted for by non-trading firms is not plotted, but can be inferred as the difference between 100% and the height of the stacked bar. See Figure 1 for definitions of trade engagement categories.

Figure 3: The Trade Status Composition of Employment, Payroll, Revenue and Exports/Imports Within Each Quartile of Export and Import Intensity



Source: LBD-Rev/LFTTD 2007-2017. Notes: The left panels of Figure 3 show the shares of firms, employment, revenue, payroll, and exports within each export intensity quartile accounted for by X, XM, and $RP\,XM$ firms, respectively. The right panels show the shares of firms, employment, revenue, payroll, and total imports within each import intensity quartile accounted for by M, XM, and $RP\,XM$ firms.

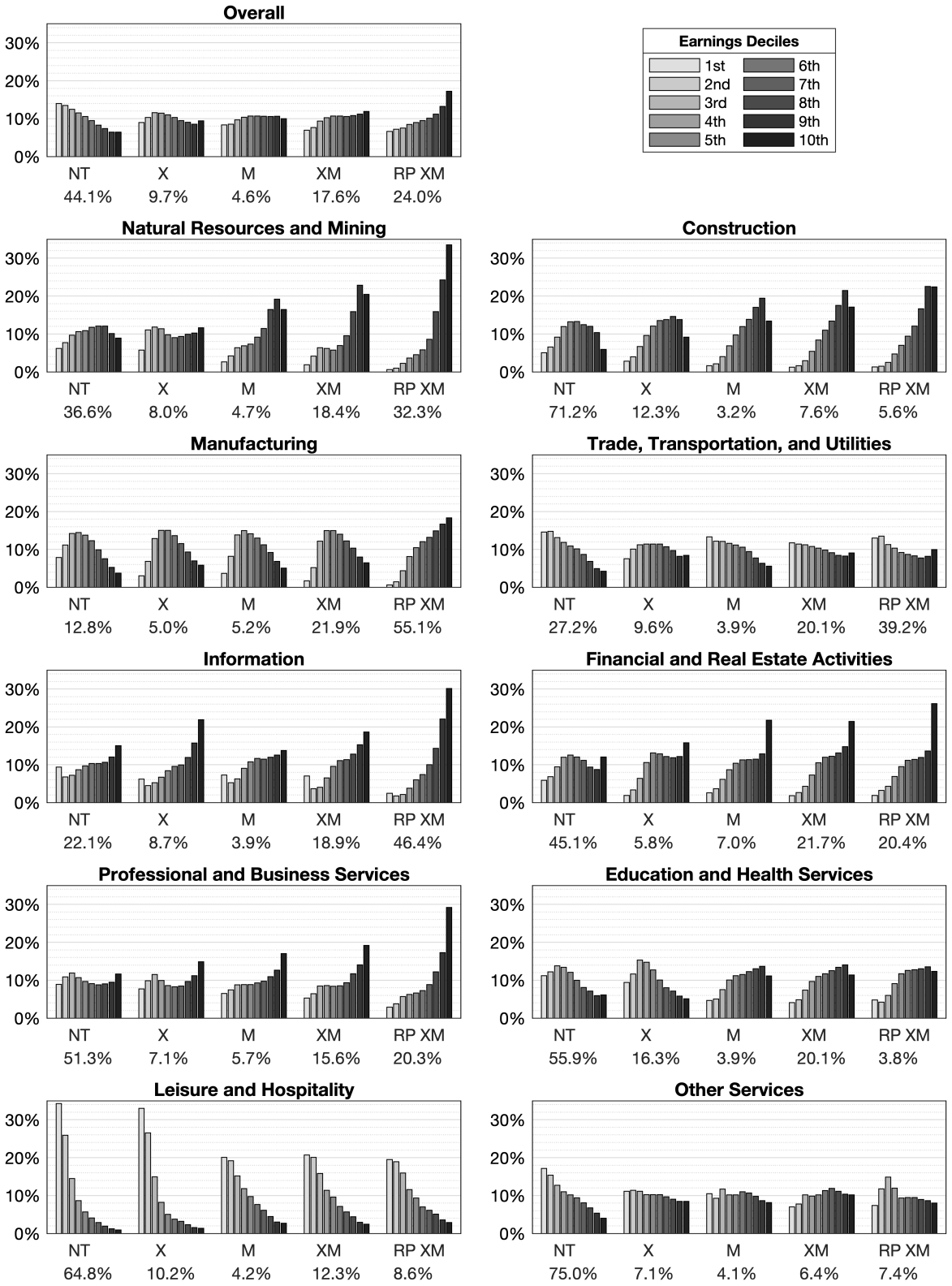
Figure 4: The Evolution of Firms' Trade Engagement Status, Weighted by Employment



Source: LBD/LFTTD 2007-2017

Notes: The top panel of Figure 4 displays, within each combination of trade engagement category and level of tenure within the category, the share of employment that is concentrated among firms that remain in the same category the following year, thus illustrating how the degree of duration dependence within a status varies across engagement status categories. The second panel displays the cross-sectional distribution of employment across tenure categories within each trade engagement status. It shows that greater persistence within $RPXM$ firms leads to a high concentration of $RPXM$ employment among firms with considerable $RPXM$ experience. The last panel shows the employment-weighted distribution of destination trade engagement status among status-switching, by origin engagement category.

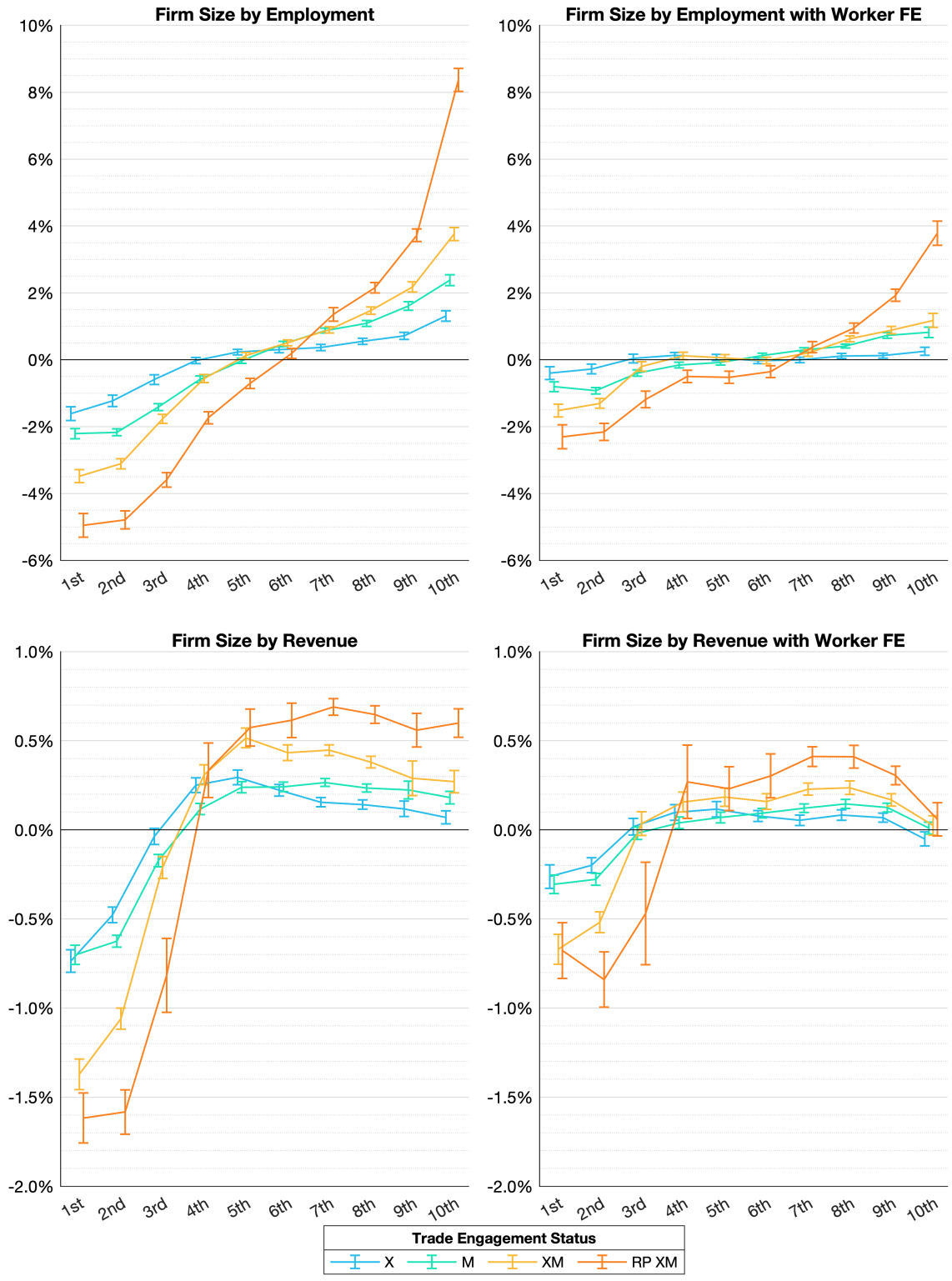
Figure 5: The Distribution of Worker Earnings Deciles within Each Trade Engagement Categories, Overall and by Industry Supersector



Source: LEHD/LFTTD 2007-2014

Notes: The top panel of Figure 5 displays the distribution of annualized earnings deciles among worker-years within firms in each trade engagement category. The remaining panels display the analogous decile distributions separately by industry. Annualized earnings are constructed by taking the mean earnings at a worker's highest earnings job int eh chosen year among all full quarters and multiplying by four to approximate the earnings the worker would receive in a full year of work. Decile cutoffs are computed separately for each year, then decile assignments are pooled among all worker-years observed between 2007 and 2014 to form the presented distributions.

Figure 6: Gaps in the Conditional Probability of Being Paid in each Earnings Decile by Trade Engagement Status Relative to Workers at Non-Trading Firms

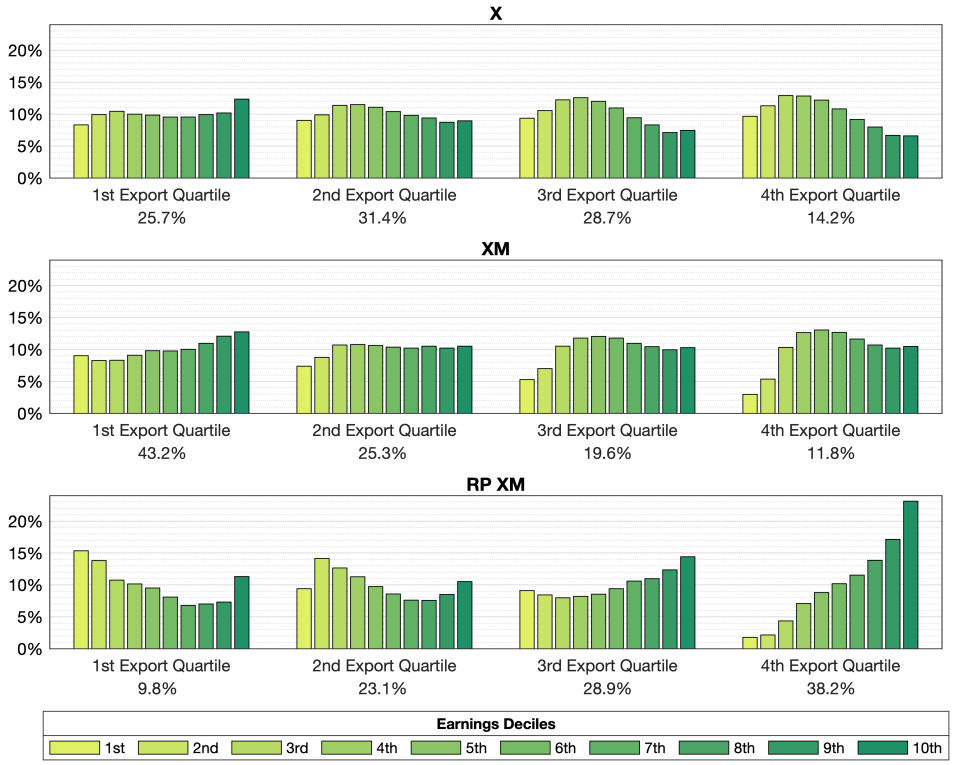


Source: LEHD/LFTTD 2007-2014

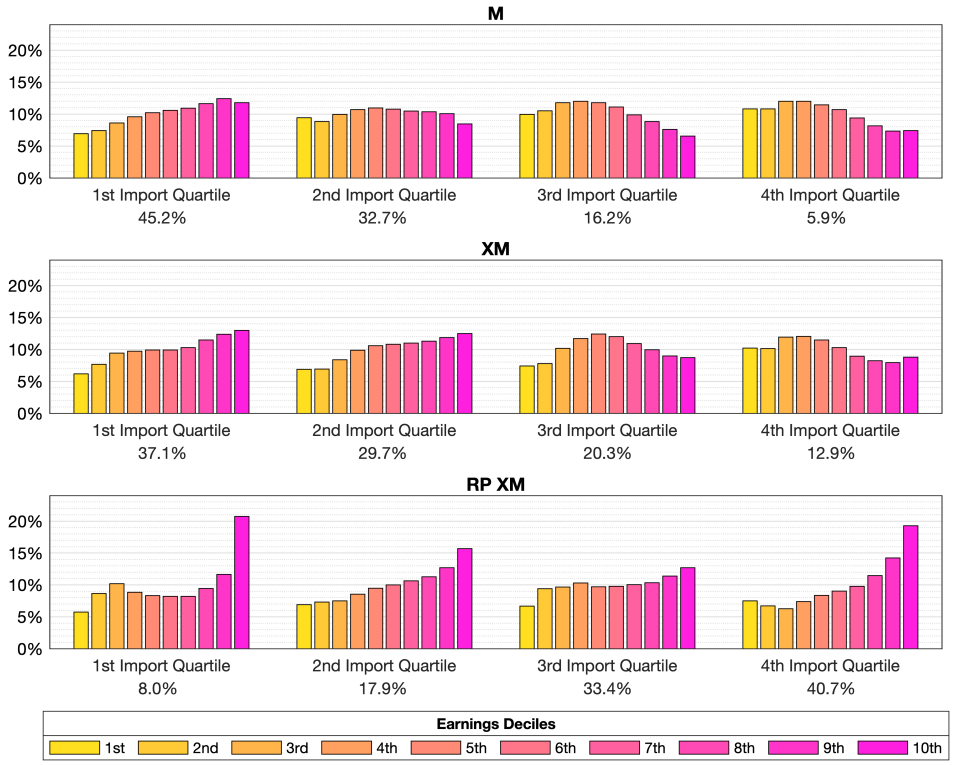
Notes: The plotted values in Figure 6 display the estimated residual gaps in the probability of receiving annual earnings in each decile of the national annualized earnings distribution among workers in each trade engagement status relative to workers at non-trading firms. The top left (right) panel residualizes with respect to indicators for firm employment (revenue) decile × supersector combinations. The bottom panels residualize with respect to worker fixed effects in addition to firm employment (or revenue) decile × supersector combinations. See Figure 5 for details about how workers are assigned to earnings deciles.

Figure 7: The Distributions of Worker Earnings Deciles by Trade Engagement Category and Export or Import Intensity Quartile

(a) **Export Trade Intensity**

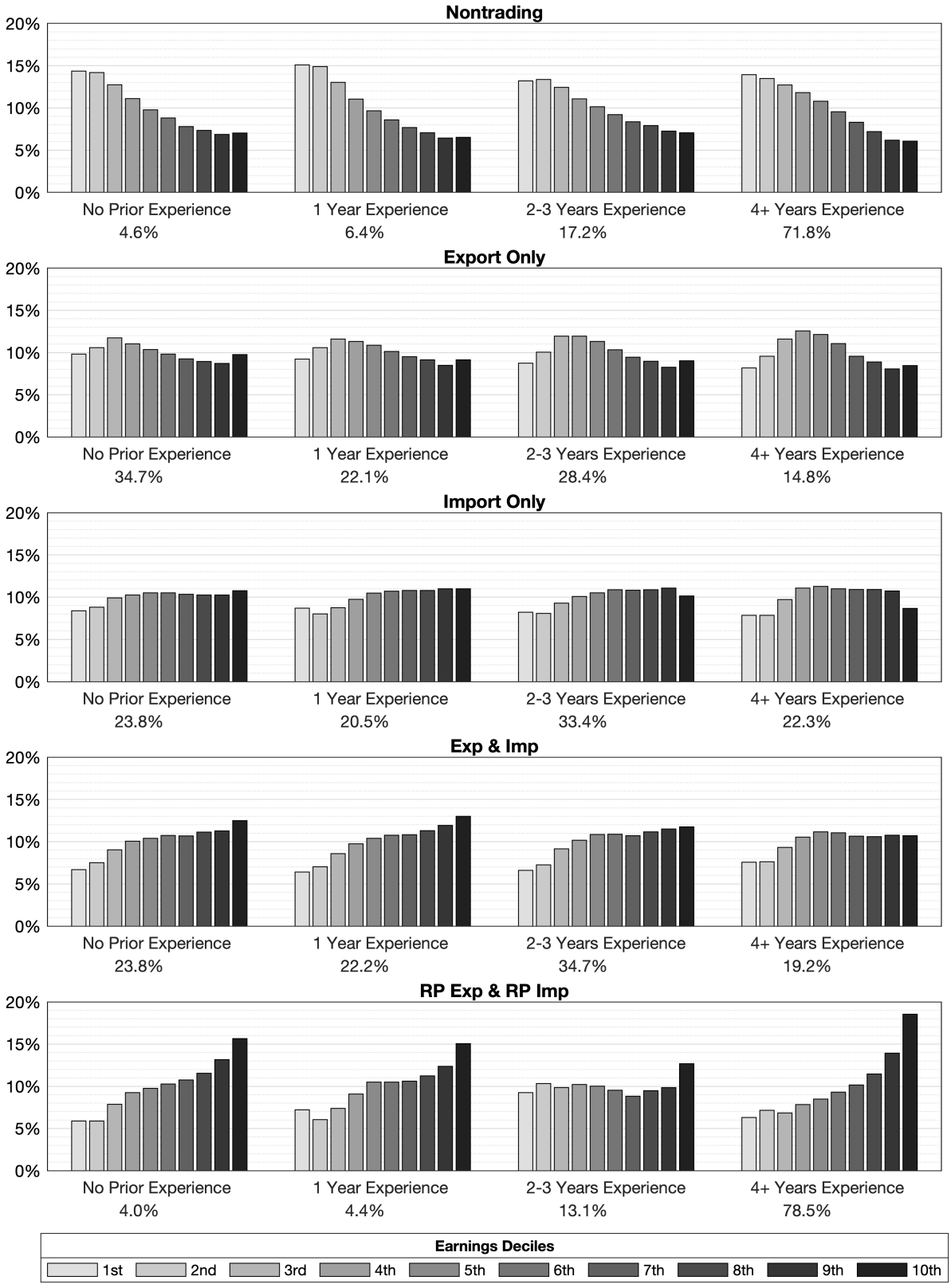


(b) **Import Trade Intensity**



Source: LEHD/LFTTD 2007-2014
Notes: The panels in Figure 7a display the distribution of annualized earnings deciles among worker-years within firms in each combination of export intensity quartile and trade engagement category among categories that engage in exporting: X, XM, and $RPXM$. The panels in Figure 7b display the corresponding decile distributions for combinations of import intensity quartile and trade engageent category for the importing categories: M, XM, and $RPXM$. See Figure 5 for details about how workers are assigned to earnings deciles.

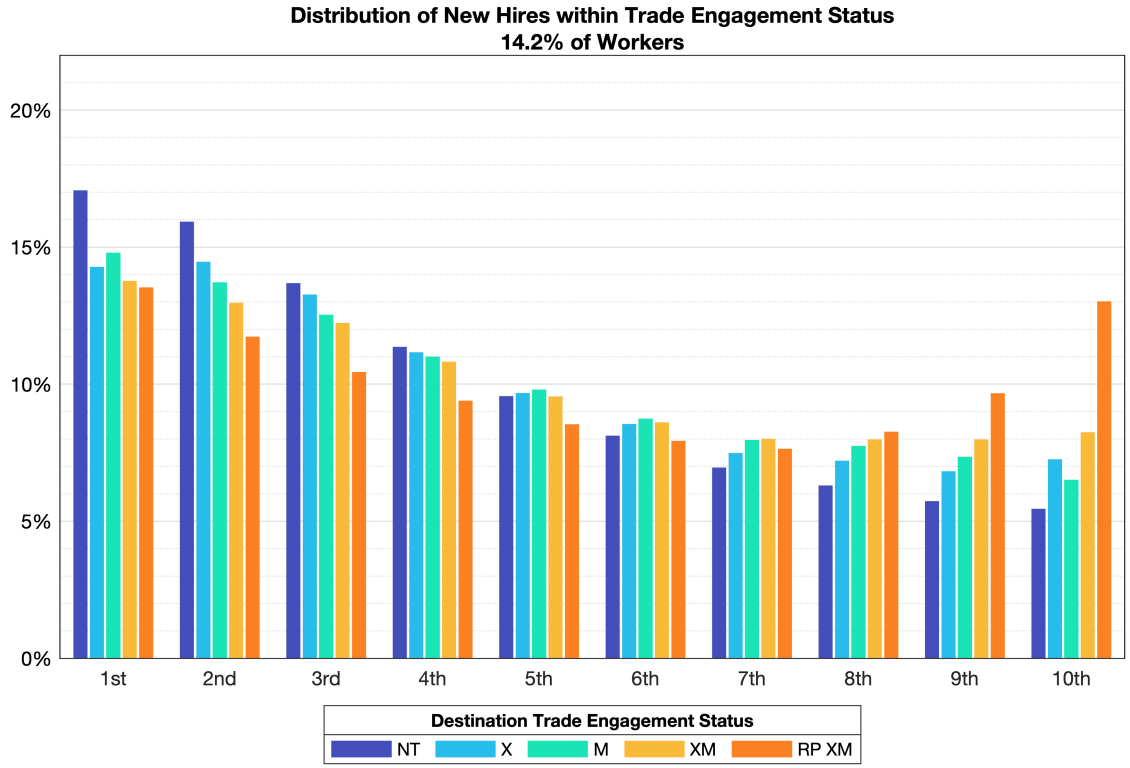
Figure 8: The Distributions of Worker Earnings Deciles by Category of Cumulative Firm Experience within Each Trade Engagement Status



Source: LEHD/LFTTD 2007-2014

Notes: The panels in Figure 7a display the distribution of annualized earnings deciles among worker-years within firms in each combination of current firm trade engagement status and firm category of cumulative experience/tenure within their current engagement status. See Figure 5 for details about how workers are assigned to earnings deciles.

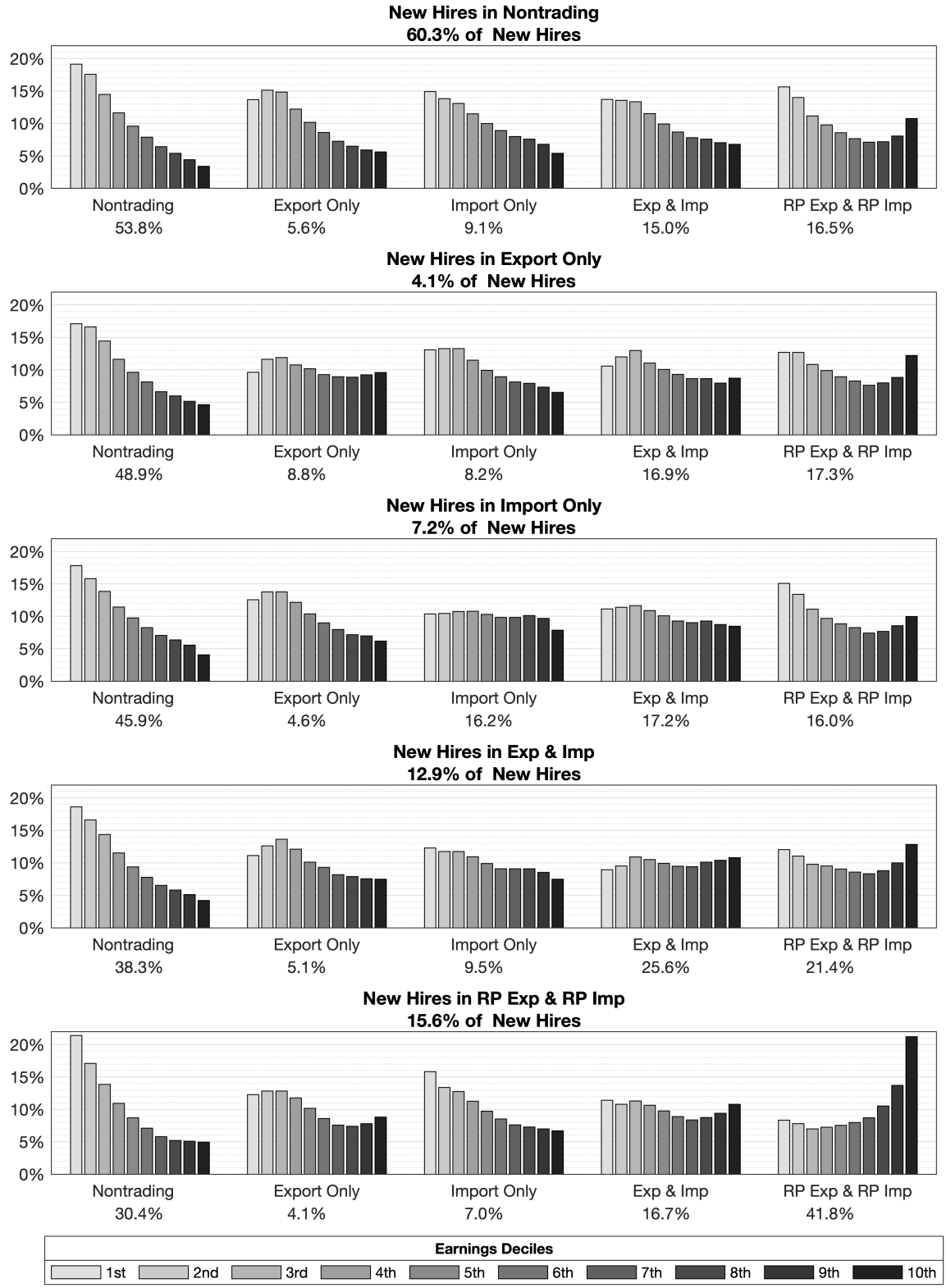
Figure 9: The Earnings Composition of Poached Workers by Trade Engagement Status



Source: LEHD/LFTTD 2007-2014

Notes: Figure 9 displays the shares of new hires made by firms in each trade engagement status whose annualized earnings in the prior year placed them in each national earnings decile. The figure demonstrates that $RP\,XM$ firms are particularly likely to poach high paid workers from other firms.

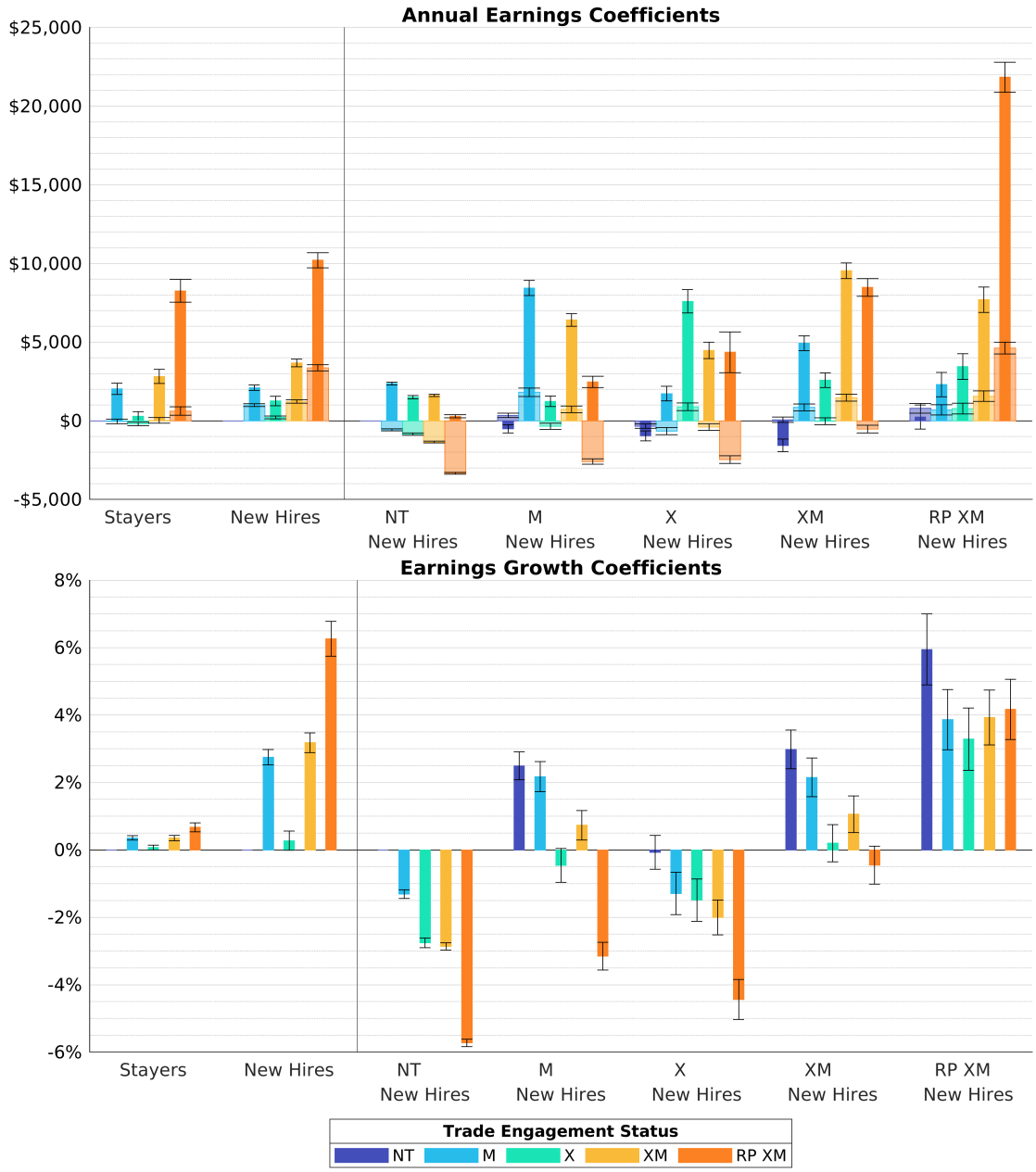
Figure 10: Poaching Patterns: Prior Year Earnings Distributions Among New Hires by Combination of Origin and Destination Trade Engagement Status



Source: LEHD/LFTTD 2007-2014

Notes: Figure 10 displays the prior year earnings decile distribution of new hires made by firms in each trade engagement status (panels) by the engagement status of the worker's prior year employer (groups within each panel). The share of the destination status' new hires that were previously working in each origin status is printed below the origin status label. See Figure 5 for details about how workers are assigned to annualized earnings deciles.

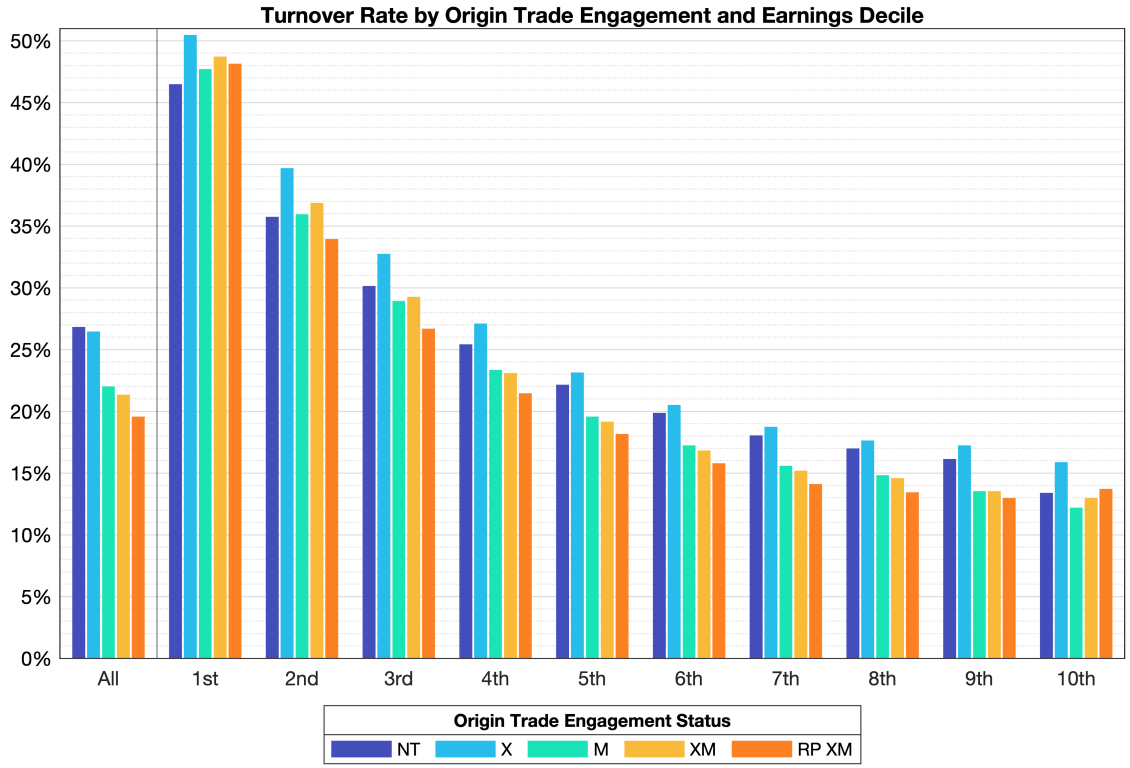
Figure 11: Higher Initial Earnings or Faster Raises? Separate Conditional Gaps in Current Earnings and Earnings Growth for Retained Workers and New Hires by Trade Engagement Status



Source: LEHD/LFTTD 2007-2014

Notes: The bars left of the vertical line in Figure 11 display coefficients capturing residual gaps in current earnings levels (first panel) or percentage growth (second panel) among retained workers (i.e. job stayers) and new hires by trade engagement status relative to those staying at or hired by non-trading firms. The bars right of the vertical line display coefficients capturing residual gaps in current earnings levels or earnings growth among newly hired workers by combination of origin (bars) and destination (labels) trade engagement status relative to newly hired non-trading workers coming from other non-trading firms. The thin bars in the first panel are based on a model that residualizes using only indicators for combinations of firm revenue decile and supersector, while the model underlying the wide bars also includes worker fixed effects in the conditioning set. The second panel controls for firm revenue decile × supersector combinations as well as indicators for earnings deciles in the prior year. Note that the reference category for the two groups of bars left of the vertical line are all job stayers and new hires, respectively, at non-trading firms, while the reference category for the groups of bars to the right of the vertical line are non-trading new hires from other non-trading firms.

Figure 12: Worker Turnover Rates for Each Trade Engagement Status - Overall and by Worker Earnings Decile



Source: LEHD/LFTTD 2007-2014

Notes: Figure 12 displays the worker turnover rate (share of previous year's workers who switch to a new primary employer) for each trade engagement status, both overall and among workers paid in each national earnings decile in the previous year. See Figure 5 for details about how workers are assigned to annualized earnings deciles.